ASSIGNMENT DEADLINE REMINDER
PULSE HACKATHON PROJECT (AUTOMATION)

OBJECTIVE:
A system that tracks assignment deadlines and sends automated reminders to students.

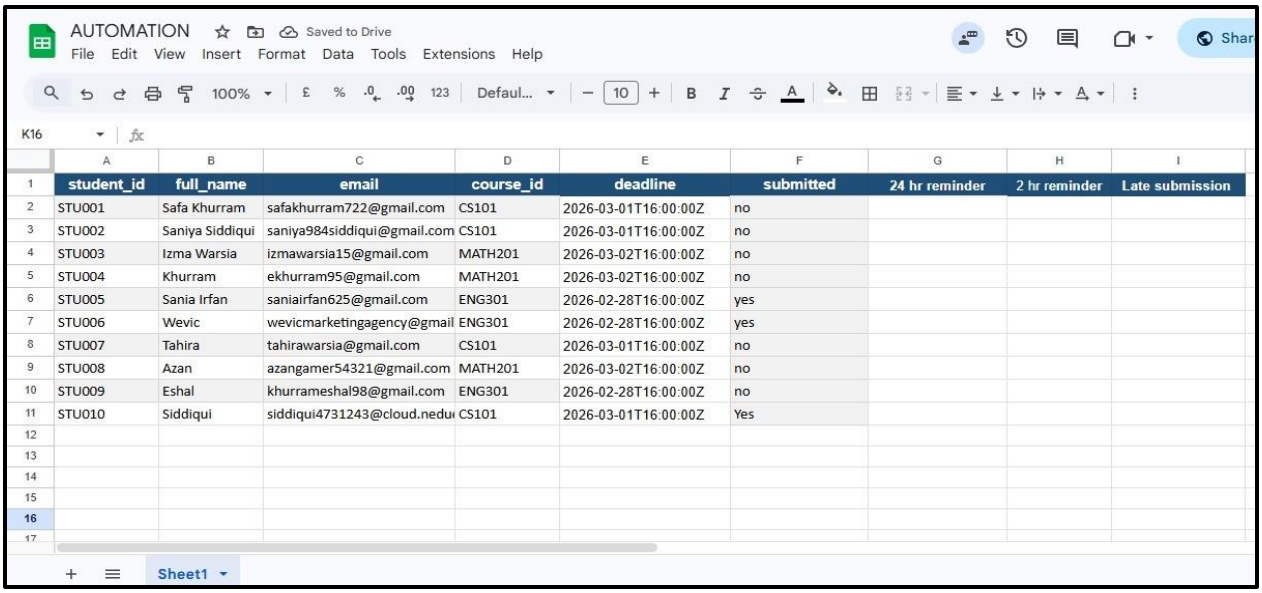
PROBLEM STATEMENT:
Students frequently miss assignment deadlines due to lack of structured reminders. Manual tracking by instructors is inefficient and error-prone.

Target Users:
Universities
Online learning platforms
LMS providers
Coaching institutes

TECH STACK:
n8n (Workflow Automation)
Google Sheets
Gmail
JavaScript

DATA STRUCTURE:

Google Sheet Columns::
Student ID
Student name
Student email
Course ID
Deadline (DateTime)
Submitted (Yes/No)
24h Reminder Sent (Yes/No)
2h Reminder Sent (Yes/No)
Late Submission (Yes/No)



student_id	full_name	email	course_id	deadline	submitted	24 hr reminder	2 hr reminder	Late submission
STU001	Safa Khurram	safakhurram722@gmail.com	CS101	2026-03-01T16:00:00Z	no			
STU002	Saniya Siddiqui	saniya984siddiqui@gmail.com	CS101	2026-03-01T16:00:00Z	no			
STU003	Izma Warsia	izmawarsia15@gmail.com	MATH201	2026-03-02T16:00:00Z	no			
STU004	Khurram	ekhurram95@gmail.com	MATH201	2026-03-02T16:00:00Z	no			
STU005	Sania Irfan	saniairfan625@gmail.com	ENG301	2026-02-28T16:00:00Z	yes			
STU006	Wevic	wevicmarketingagency@gmail	ENG301	2026-02-28T16:00:00Z	yes			
STU007	Tahira	tahirawarsia@gmail.com	CS101	2026-03-01T16:00:00Z	no			
STU008	Azan	azangamer54321@gmail.com	MATH201	2026-03-02T16:00:00Z	no			
STU009	Eshal	khurrameshal98@gmail.com	ENG301	2026-02-28T16:00:00Z	no			
STU010	Siddiqui	siddiqui4731243@cloud.nedu	CS101	2026-03-01T16:00:00Z	Yes			

<u>SOLUTION</u>:

SYSTEM DESIGN:

The system uses a scheduled automation workflow that runs every 15 minutes and evaluates assignment deadlines using conditional logic.

WORKFLOW:

The workflow follow following process :

1. **Schedule Trigger**

Runs the workflow every 15 minutes to check assignment deadlines.

2. **Get Row(s) in Sheet**

Reads all assignment entries from the connected Google Sheet.

3. **If Node (Check Assignment Submitted / Reminder Sent for 24h)**

Checks if the assignment is not submitted and 24-hour reminder has not been sent.

4. **Send a Message (24-Hour Reminder)**

Sends an email reminder for assignments with less than 24 hours left.

5. **Update Row in Sheet (24-Hour Reminder**)

Updates Reminder Sent column to Yes after sending the 24-hour reminder.

6. If **Node (Check Assignment Submitted / Reminder Sent for 2h**)

Checks if the assignment is not submitted and 2-hour reminder has not been sent.

7. **Send a Message (2-Hour Reminder)**

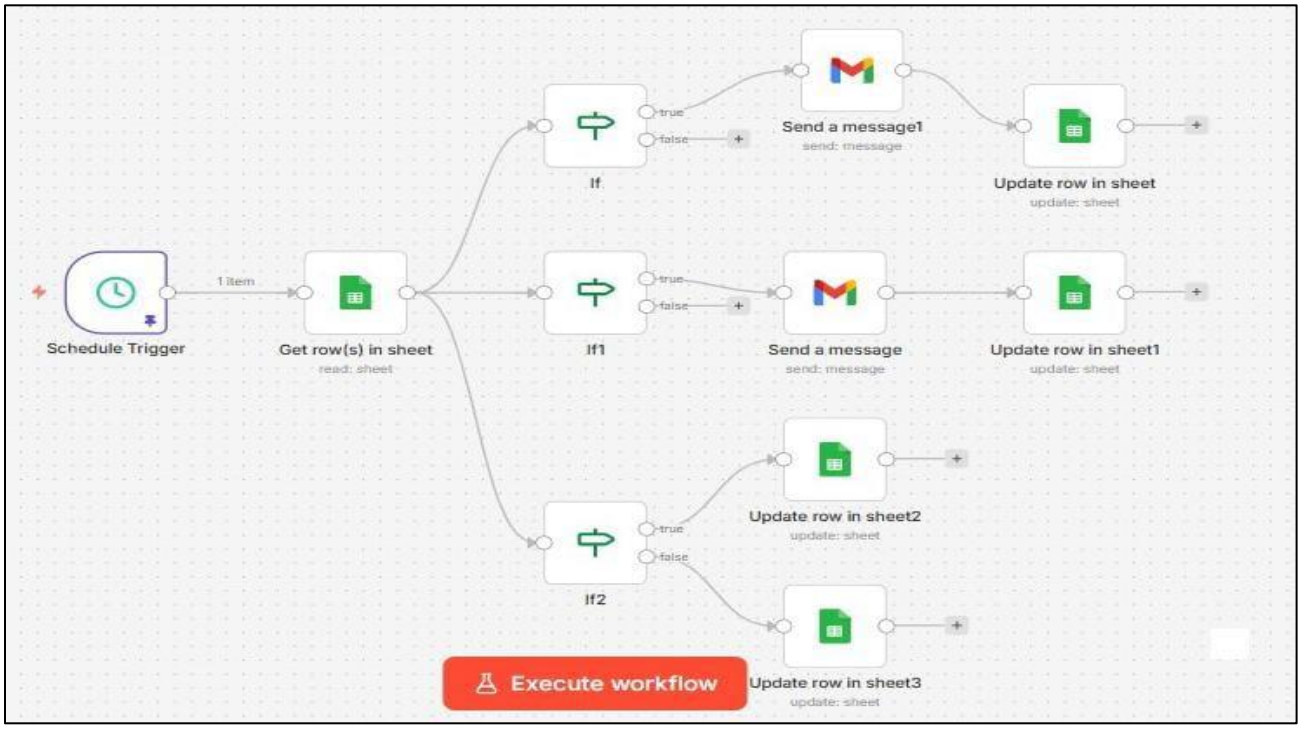
Sends an email reminder for assignments with less than 2 hours left.

8. **Update Row in Sheet (2-Hour Reminder)**
Updates Reminder Sent 2h column to Yes after sending the 2-hour reminder.

9**. If Node (Check Late Submission)**
Checks if current time > assignment deadline.

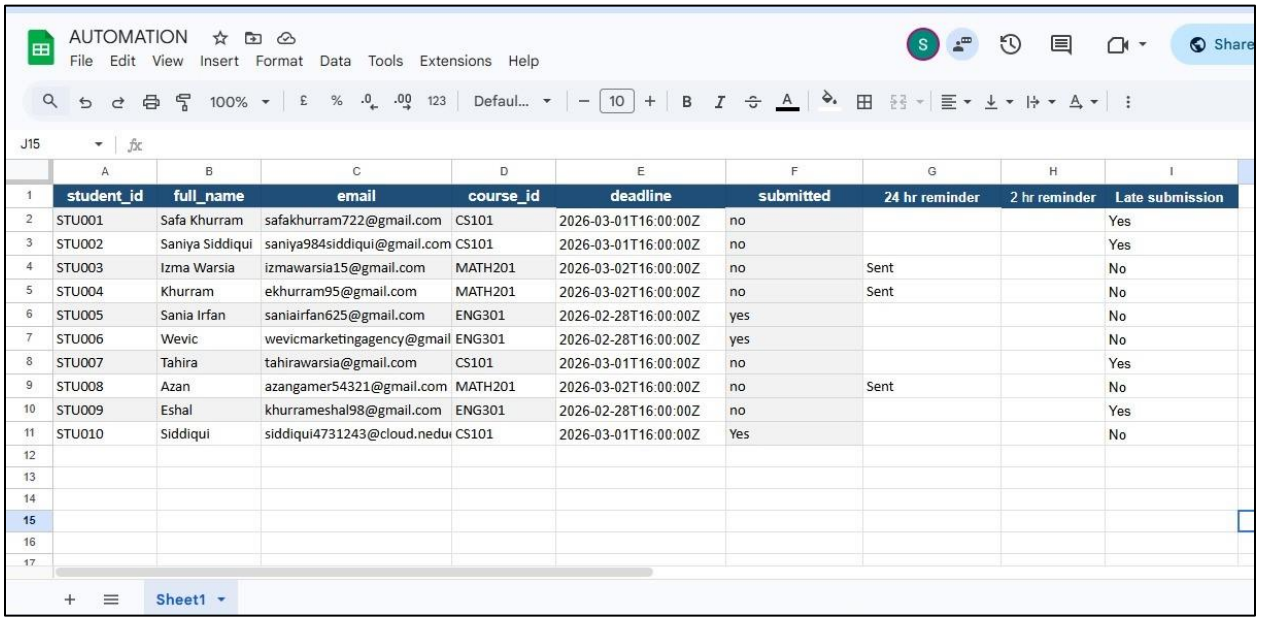
10. **Update Row in Sheet (Late Submission)**
Marks Late Submission column as Yes if overdue, No if still pending.

<u>BEFORE EXECUTION :</u>



AFTER EXECUTION:

After successful execution of workflow, the record is updated on google sheets:



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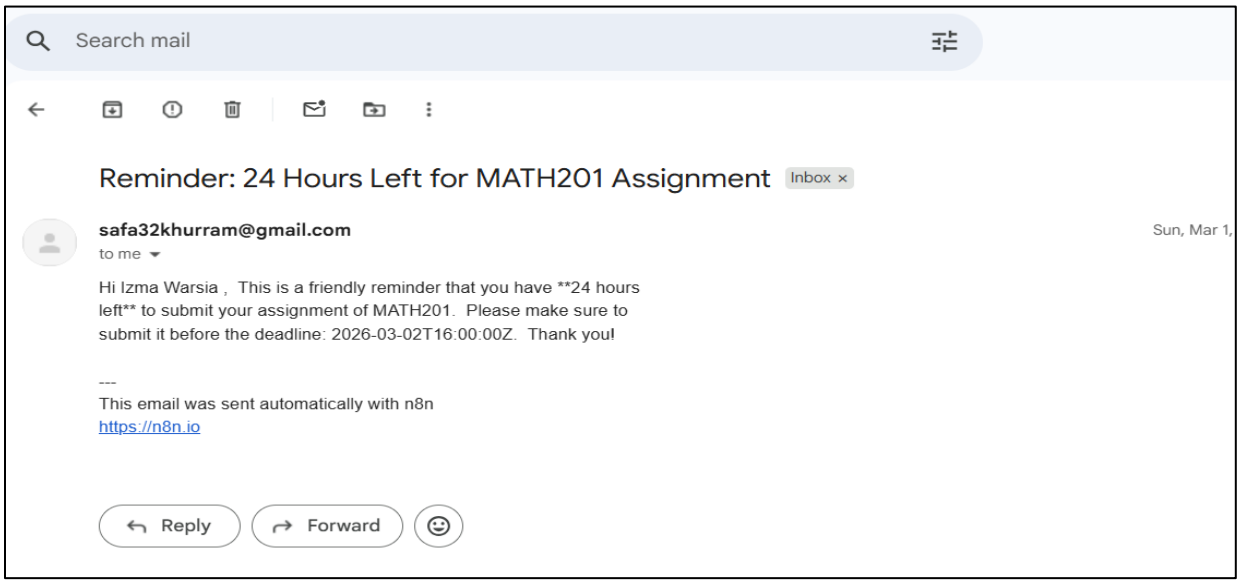
STU002	Saniya Siddiqui	saniya984siddiqui@gmail.com	CS101	2026-03-01T16:00:00Z	no			Yes
STU003	Izma Warsia	izmawarsia15@gmail.com	MATH201	2026-03-02T16:00:00Z	no	Sent		No
STU004	Khurram	ekhurram95@gmail.com	MATH201	2026-03-02T16:00:00Z	no	Sent		No
STU005	Sania Irfan	saniairfan625@gmail.com	ENG301	2026-02-28T16:00:00Z	yes			No
STU006	Wevic	wevicmarketingagency@gmail	ENG301	2026-02-28T16:00:00Z	yes			No
STU007	Tahira	tahirawarsia@gmail.com	CS101	2026-03-01T16:00:00Z	no			Yes
STU008	Azan	azangamer54321@gmail.com	MATH201	2026-03-02T16:00:00Z	no	Sent		No
STU009	Eshal	khurrameshal98@gmail.com	ENG301	2026-02-28T16:00:00Z	no			Yes
STU010	Siddiqui	siddiqui4731243@cloud.nedu	CS101	2026-03-01T16:00:00Z	Yes			No

REMINDER MESSAGE RECEIVED AT GMAIL:



CORE LOGIC SUMMARY:

If deadline − current_time ≤ 24h AND reminder not sent → Send 24h reminder
If deadline − current_time ≤ 2h AND reminder not sent → Send 2h reminder
If current_time > deadline AND not submitted → Mark as Late

STATE MANAGEMENT:

The system prevents duplicate reminders by updating reminder status columns after each email is sent. Each reminder type (24h / 2h) is tracked independently.

MVP (Minimum Viable Product)

The MVP demonstrates the core functionality of automated assignment reminders:

Implement features:

Scheduled Trigger : Workflow runs every 15 minutes to check deadlines.
24-Hour Reminder :Emails students with assignments due in the next 24 hours; marks reminder as sent.
2-Hour Reminder : Emails students with assignments due in 2 hours; marks reminder as sent.
Late Submission Tracking :Marks assignments as “Late” if past the deadline.
Google Sheets Integration :Stores all assignment data and reminder statuses.

GROUP MEMBERS:

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